

MultiRole Hiring Automation System

Walkthrough Document

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Trilles AI n8n Automation Developer Assignment

1. Project Overview

This document explains the architecture, decisions, assumptions, and known limitations of the MultiRole Hiring Automation System built in n8n. The system fully automates the hiring pipeline for two roles: Software Engineer (SWE) and Business Development Manager (BDM), from application ingestion to candidate communication.

2. Workflow Architecture

The system is split into two separate n8n workflows as required by the assignment.

Workflow 1: Main Processing Pipeline

Runs on a schedule every 2 hours. Handles all data processing from ingestion to storage.

- Schedule Trigger fires every 2 hours
- Reads new applications from both SWE and BDM Google Sheets in parallel
- Merges both streams into one unified flow using the Merge node in Append mode
- Filters out already processed candidates using the Processed column
- Exits gracefully if no new applications are found
- Splits into batches of 1 to process candidates one at a time
- Cleans and standardizes data from both sheets into a unified format
- Detects role by checking for rolespecific fields (Tech Stack = SWE, Industry Background = BDM)
- Checks if a resume was uploaded and routes accordingly
- Downloads the resume file from Google Drive using the file ID extracted from the Drive URL
- Extracts text content from the PDF using the Extract from File node
- Merges extracted resume text back with candidate data
- Builds a rolespecific AI prompt with full resume context
- Sends the prompt to Mistral AI via HTTP Request and receives a structured JSON evaluation
- Parses the AI response to extract score, classification, summary, strengths, concerns
- Appends the full candidate record to the Master Candidates sheet
- Updates the source sheet (SWE or BDM) to mark the candidate as Processed

Workflow 2: Email Dispatcher

Runs on a separate schedule every 30 minutes. Handles all candidate communication independently.

- Schedule Trigger fires every 30 minutes
- Reads the Master Candidates sheet and filters for candidates where Email Sent = No
- Exits gracefully if no pending candidates are found
- Splits into batches of 1 to process emails one at a time
- Routes each candidate to the correct email branch using a Switch node based on classification
- Strong candidates receive an HTML interview invitation email and a Google Calendar event is created with them added as an attendee
- Average candidates trigger a detailed notification email to the hiring manager including the AI score, summary, strengths, and concerns
- Weak candidates receive a professional and warm rejection email encouraging future applications
- After each email is sent the Master Sheet is updated: Email Sent = Yes and Status is updated accordingly

3. Technology Stack

Component	Choice
Automation Platform	n8n (local installation via npm)
Application Forms	Google Forms (one per role)
Data Storage	Google Sheets (SWE, BDM, and Master sheets)
Resume Storage	Google Drive (via Google Forms file upload)
AI Model	Mistral AI (mistralsmalllatest) via REST API
Email Service	Gmail via OAuth2
Calendar	Google Calendar via OAuth2
PDF Extraction	n8n builtin Extract from File node

4. AI Screening Design

The AI screening is rolespecific. Each role uses a completely different evaluation prompt with different criteria.

SWE Evaluation Criteria

- Programming languages and frameworks
- Project complexity and realworld impact
- GitHub activity and open source contributions
- System design and architecture experience

- Overall technical depth

BDM Evaluation Criteria

- Sales and revenue track record
- Client relationship management
- Market expansion and business strategy experience
- Communication and leadership skills
- Industry network and business acumen

Scoring and Classification

The AI returns a score from 0 to 100 and classifies candidates as strong, average, or weak using these thresholds:

- Strong: score 60 or above
- Average: score 35 to 59
- Weak: score below 35

Prompts instruct the AI to be generous with early career candidates and evaluate based on their experience level rather than holding junior candidates to senior standards.

5. Key Design Decisions

Two Separate Workflows

Email sending is intentionally separated from main processing as required by the assignment. This makes the system more resilient. If email sending fails it does not affect resume processing, and vice versa.

Google Forms as Entry Points

Google Forms were used because they automatically dump data into Google Sheets and handle file uploads to Google Drive. This eliminates the need for a custom frontend and keeps everything within the Google ecosystem.

Role Detection by Field Presence

Instead of using a separate column to track role, the system detects the role by checking which fields are present. If Tech Stack exists it is SWE. If Industry Background exists it is BDM. This works reliably because the two forms have different fields.

Mistral AI over Gemini

Mistral AI was chosen after Gemini's free tier quota was exhausted during development. Mistral provides a reliable free tier with consistent JSON output which is critical for parsing the AI response correctly.

Referencing Earlier Nodes with \$()

Because several nodes in the chain (Google Drive download, Extract from File, Gmail) return their own output and drop earlier data, the Switch node name is referenced directly using the `$()` expression to retrieve candidate data reliably throughout the email workflow.

6. Assumptions

- Candidates must be signed into a Google account to upload a resume via the form. This is a Google Forms file upload limitation.
- Resume files are in PDF format. The Extract from File node is configured for PDF only.
- Both Google Forms feed into sheets named Form Responses 1 which is the default name Google assigns.
- The hiring manager email for average candidate notifications is the same Gmail account used for sending emails.
- Interview scheduling defaults to 3 days from the time of processing with a 1 hour duration.
- Candidates who did not upload a resume are logged as an edge case and skipped without breaking the loop.

7. Error Handling and Edge Cases

No New Applications

Both workflows check if there are any items to process before doing any work. If no new applications or no pending emails are found the workflow exits cleanly without errors.

Missing Resume

The IF Resume Exists node checks the `hasResume` flag. Candidates without a resume are routed to an edge case handler node that logs the skip and loops back without breaking the batch processing.

AI Response Parsing Failure

The parse code node wraps `JSON.parse` in a try/catch block. If the AI returns malformed JSON the candidate is assigned a default classification of average with a score of 0 and a note that the response could not be parsed.

Multiple Candidates Per Run

The Split in Batches node processes one candidate at a time. Both processing branches (SWE and BDM update) connect back to the loop node ensuring all candidates are processed regardless of role.

Duplicate Prevention

The Processed column in SWE and BDM sheets and the Email Sent column in the Master sheet ensure candidates are never processed or emailed twice across multiple workflow runs.

8. Known Limitations

- Mistral free tier has rate limits. Running too many candidates in quick succession may trigger temporary throttling.
- Google Calendar invites are created 3 days from the processing time. There is no logic to avoid weekends or public holidays.
- The system does not support resume formats other than PDF.
- If a candidate submits duplicate applications they will be processed twice since deduplication is based on row position not email address.
- The AI classification thresholds are fixed in the prompt. A more advanced version would allow the hiring manager to configure these thresholds dynamically.
- There is no retry mechanism if the Mistral API call fails. A failed API call will cause that candidate to be skipped in the current run.

9. What I Would Improve With More Time

- Add a Slack or WhatsApp notification to the hiring manager in real time when a strong candidate is found
- Build a simple dashboard in Google Sheets using charts to visualize pipeline stats by role and classification
- Add email deduplication to prevent processing the same candidate twice across runs
- Implement retry logic for the AI API call with exponential backoff
- Allow interview scheduling to consider working days only and let the hiring manager set available time slots
- Add a feedback loop where the hiring manager can override the AI classification directly in the sheet and trigger a reemail

10. Walkthrough Loom Video Link

Link: <https://www.loom.com/share/b59e6dc089304364b9f6973bd231e29c>